



EXPLAINABLE E-COMMERCE DECISION INTELLIGENCE

Make the next product decision explainable

A four-week pilot proposal for marketplace merchandising teams

TEAM YOUNGHTT

AREA 303 · TRACK 1 · TOPIC 05



OUR PROPOSAL

Approve a four-week Indonesia pilot of SKUNIVO

Use one weekly merchandising meeting to prove faster, more consistent and more accountable product triage.

THE DECISION

Turn 475 Indonesia listings into one auditable priority queue.

83

active-review products

10

shops in pilot scope

4

platform categories

WHO USES IT

- Category and merchandising managers
- Promotion planners
- Commercial analytics teams

WHAT THEY RECEIVE

A ranked queue, recommendation, confidence, three reasons and a human decision log.

A weekly review cannot treat every listing as equally urgent

The user does not need another dashboard; they need a defensible order of action.

BEFORE SKUNIVO

1,157

listings compete for attention

- Which hero SKU must be protected?
- Which promotion deserves a test?
- Which deep discount should be reviewed?

PRICE

DISCOUNT

RATING

LIKES

SOLD-VALUE PROXY

SHOP CREDIBILITY

The data supports prioritization—not forecasting

The proposal is deliberately limited to decisions the current evidence can support.

1,157

latest listings

20

shops

2

markets

3

snapshot dates

NOT AVAILABLE

Orders • Customers • Costs • Inventory history • Ad spend • Realized revenue • Treatment/control

No demand forecast, causal promotion lift, profit optimization or ROAS claim.

AVAILABLE

- Price and displayed promotion
- Ratings, volume and likes
- Shop and category context
- Sold-value proxies

Indonesia offers the clearest first decision opportunity

It contains 85% of the active-review queue and a materially deeper discount environment.

INDONESIA · PILOT NOW

83 / 98

active-review products

53%

median displayed discount

81

discount reviews

- Grouped shop validation produces a usable rank benchmark.
- The review problem is concentrated enough for a focused pilot.

VIETNAM · LEARN NEXT

15 / 98

active-review products

19%

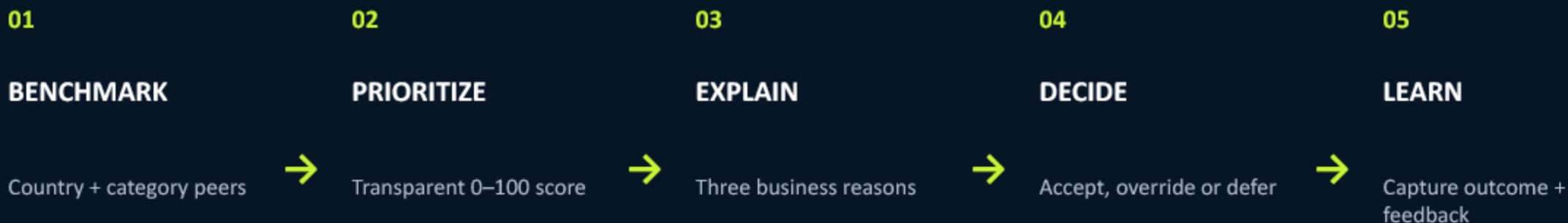
median displayed discount

- Broader category coverage makes it a valuable second phase.
- Lower shop-held-out rank quality requires cautious model confidence.

THE SOLUTION

SKUNIVO converts signals into a governed decision workflow

The AI proposes evidence; a merchant remains accountable for the action.



BUSINESS OUTPUT

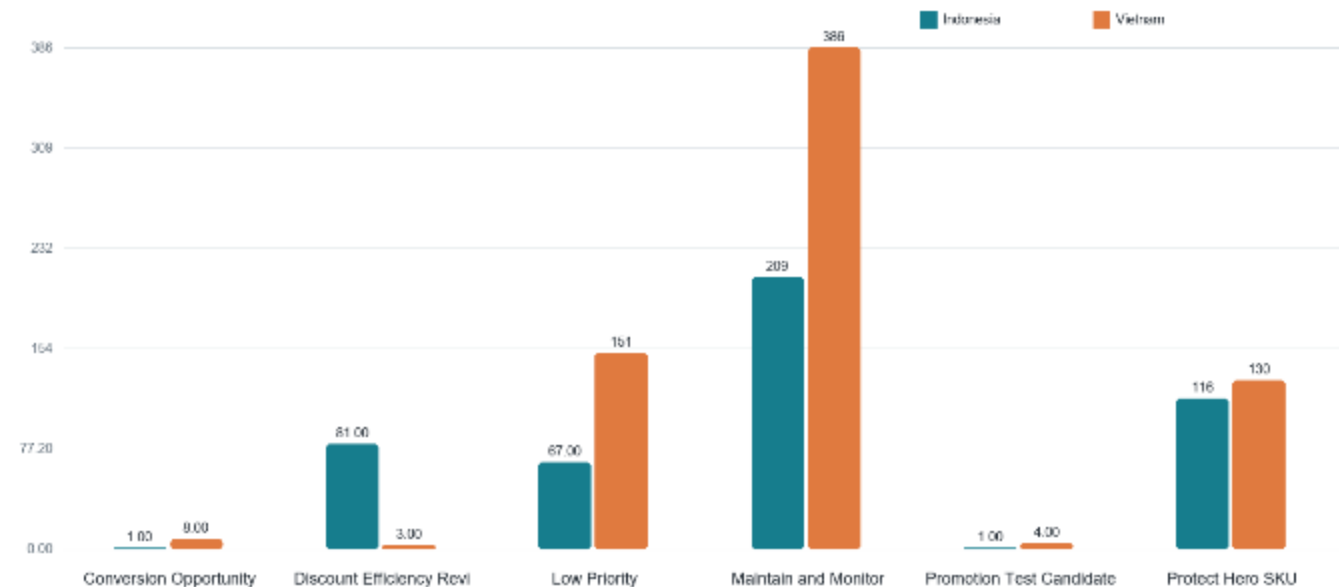
One review queue for hero protection, promotion testing and discount-efficiency decisions.

SKUNIVO concentrates active review on 8.5% of listings

The remaining portfolio stays visible but does not consume the same decision bandwidth.

Recommendation mix differs by market context

Decision-support labels assigned from score components and peer position



Labels guide review; they do not guarantee outcomes.

98

products for active review

84

discount reviews

9

conversion opportunities

5

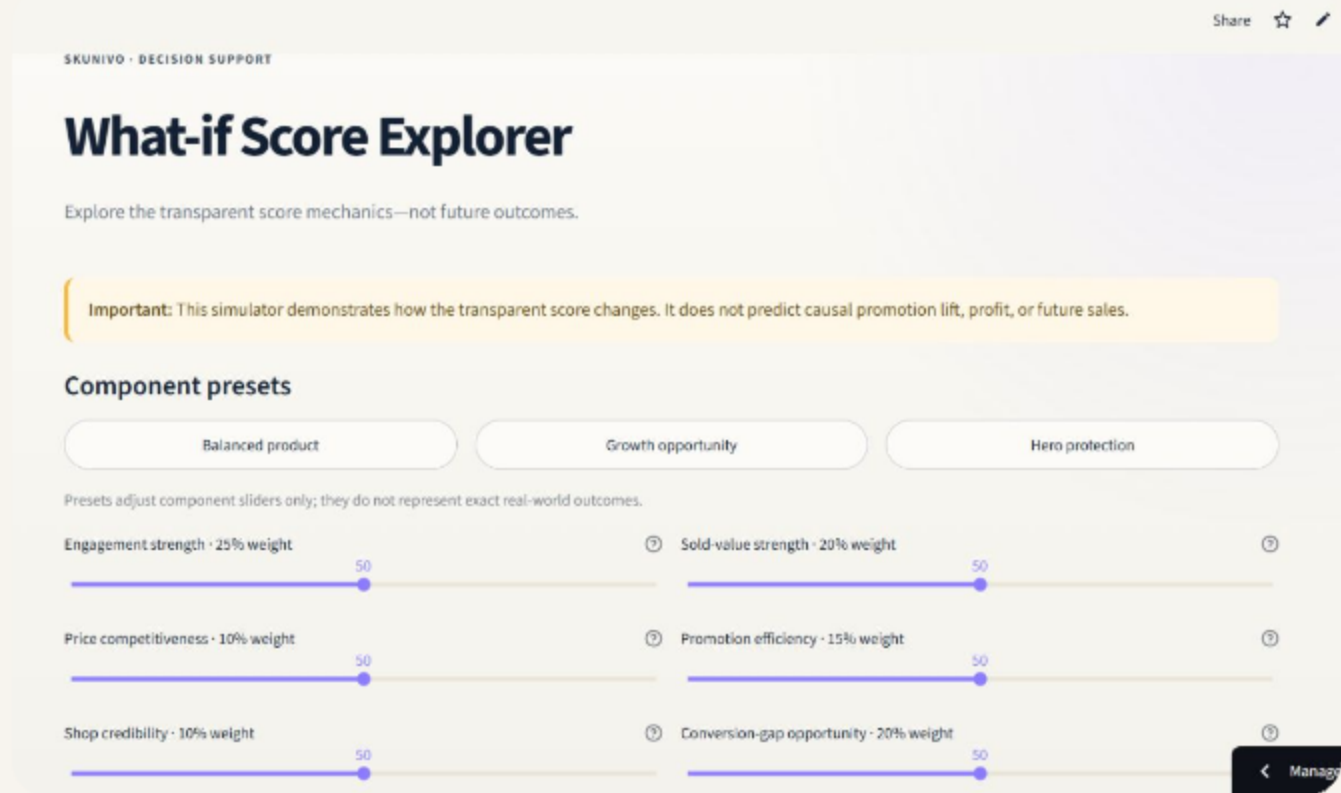
promotion test candidates

Recommendation labels guide review; they do not guarantee outcomes.

Every score can be reconstructed, challenged and reweighted

Six peer-relative components make the recommendation inspectable before any action is taken.

25%	Engagement strength
20%	Sold-value strength
10%	Price competitiveness
15%	Promotion efficiency
10%	Shop credibility
20%	Conversion-gap opportunity



A 54% discount becomes a review question

SKUNIVO does not recommend a deeper discount; it proposes a governed review of conflicting signals.

LAVOJOY OFFICIAL SHOP

Let It Glow 10% Niacinamide Sunscreen Serum

54%

displayed discount

49th

likes percentile

33rd

sold-value
percentile

Opportunity score 37.70 • Medium confidence

SKUNIVO PROPOSES

Discount Efficiency Review

INTERPRETATION

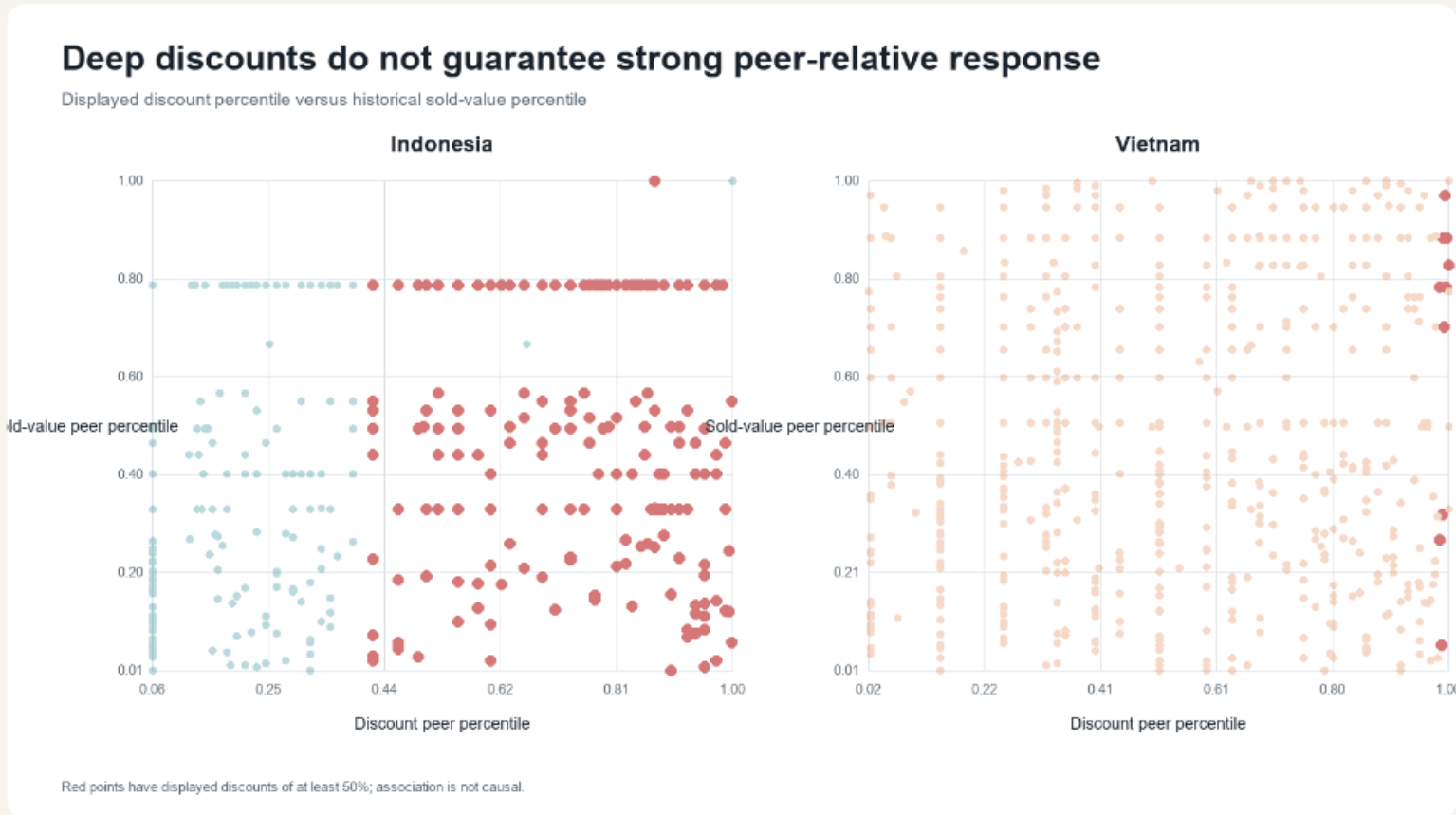
- The current discount is deep, but peer-relative response is modest.
- Do not deepen the discount before diagnosing content, availability and mechanism.

HUMAN DECISION

Approve a controlled review or override with a documented reason.

Deep discount does not guarantee strong peer-relative response

The pattern justifies a review queue; incrementality still requires an experiment.



84

discount-efficiency reviews

- 81 are in Indonesia.
- Only five products qualify as promotion test candidates.
- Causal lift is explicitly out of scope until treatment/control data exists.

AI proposes; merchants decide; overrides become learning data

The product explanation keeps evidence, confidence and business action in the same view.

Share ☆ ✎

SKUNIVO · DECISION SUPPORT

Product Explanation

Audit the evidence behind one product recommendation.

Country

Indonesia ▾

Shop

Cyecare Official Store ▾

Search by item ID or product name

Type part of a product name or an item ID

Item ID and product

28934445366 · Cyecare Hair Mask Masker Rambut Keratin Hyaluronic Acid Treatment Melembutkan Meluruskan Rambut Rusak Kering ▾

PRODUCT DECISION RECORD

Cyecare Hair Mask Masker Rambut Keratin Hyaluronic Acid Treatment Melembutkan Meluruskan Rambut Rusak Kering Mengembang 150ml

< Manan

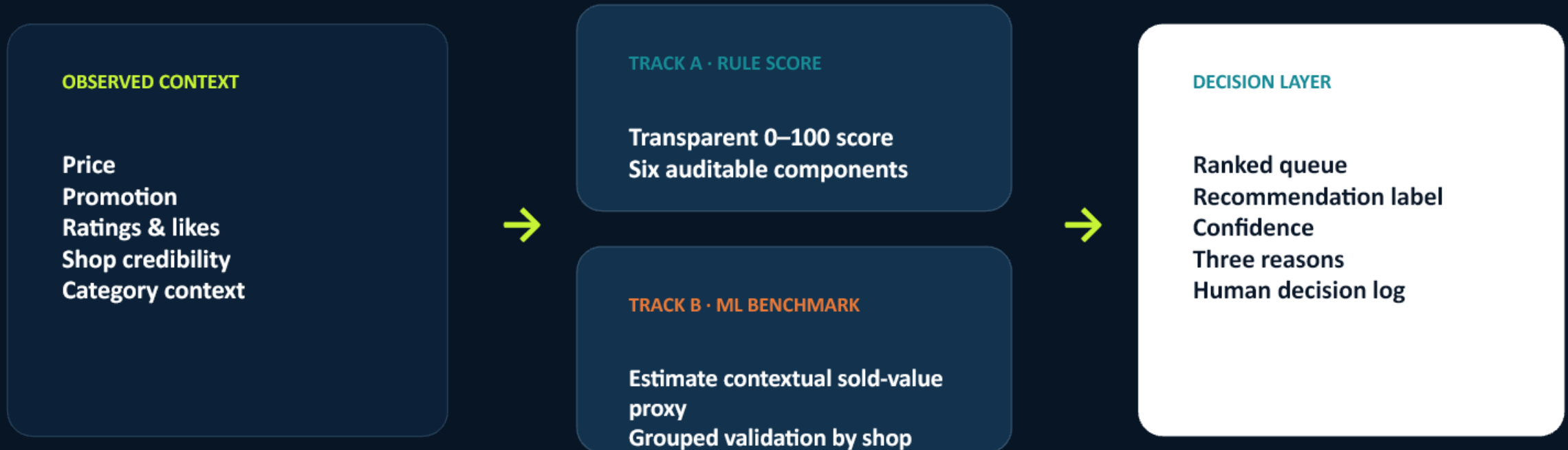
ONE DECISION RECORD

- Observed marketplace signals
- Peer percentile benchmarks
- AI contextual benchmark and confidence
- Accept, override or defer with a reason

This is decision support—not automatic execution.

Two analytical tracks prevent the model from becoming the decision

The transparent score drives the queue; ML provides a contextual second opinion.

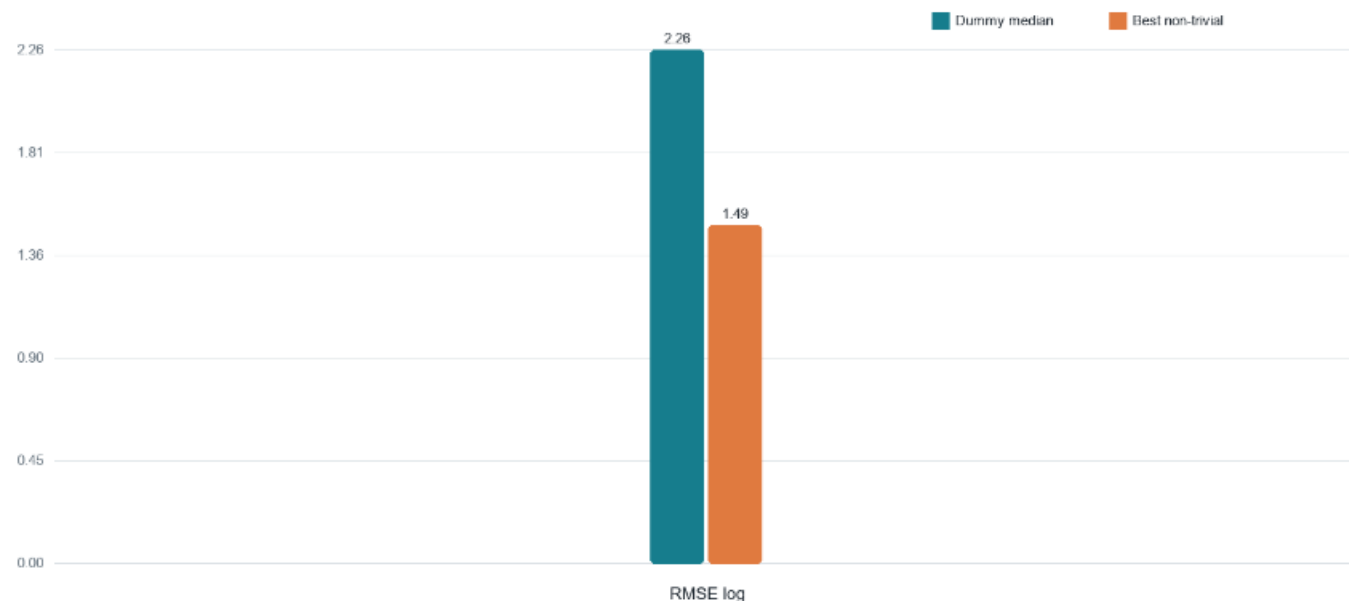


Model performance supports Indonesia-first deployment

All validation is grouped by shop; Vietnam remains visible with lower model confidence.

Non-trivial models are tested against a median baseline

Mean shop-grouped cross-validation RMSE across countries and feature experiments



Lower is better. Descriptive features are leakage-prone for future prediction and are reported separately.

0.77

Indonesia Spearman

0.35

Vietnam Spearman

Deployment rule

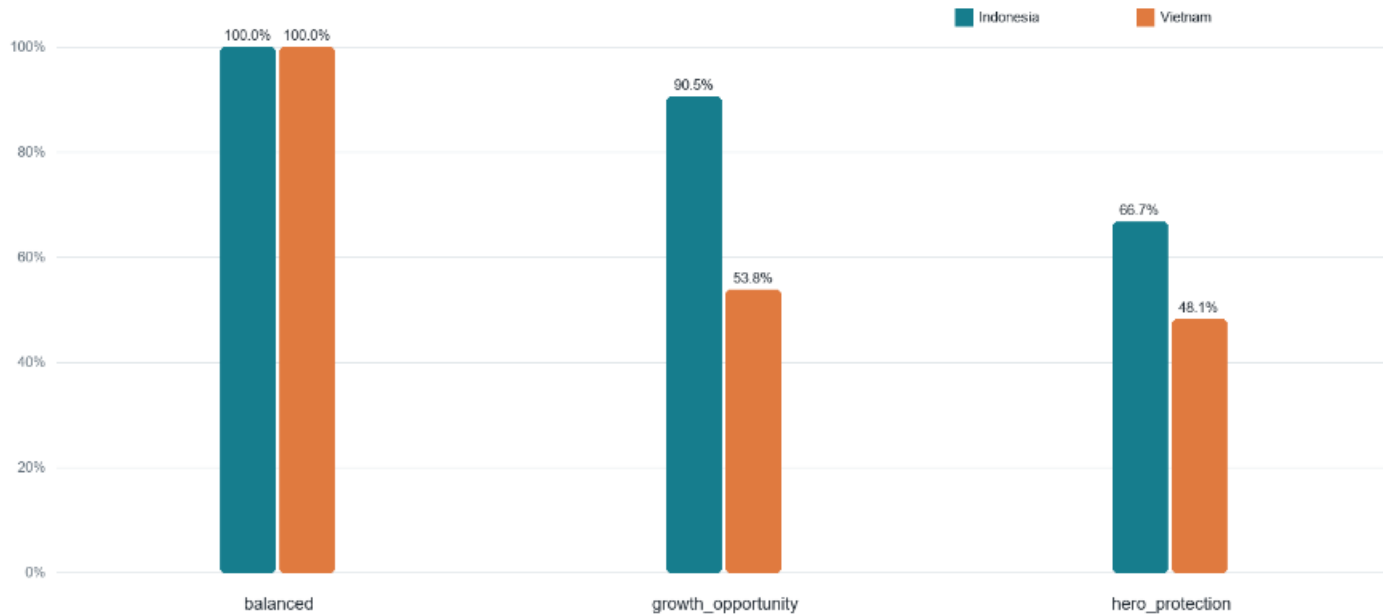
Use the model as a benchmark in Indonesia; reduce confidence and keep rule-based explanations primary in Vietnam.

The ranking remains stable under strategic reweighting

Changing reasonable weights preserves overall order while adapting the top queue.

Top-ranked products remain reasonably stable across score weights

Jaccard overlap of each configuration's top 20 with the balanced top 20



Balanced is 100% by definition; higher alternative overlap means greater ranking robustness.

>0.986

rank correlation vs. balanced

90%

Indonesia top-20 overlap
under growth weighting

Governance implication

Business owners may choose a preset, but every weight change remains visible and testable.

Operational discipline before revenue promises

The pilot measures whether better prioritization changes how decisions are made.

FASTER

Reduce time spent scanning low-priority listings.

MORE CONSISTENT

Apply the same peer logic across shops and categories.

MORE ACCOUNTABLE

Record the evidence, owner and reason behind each decision.

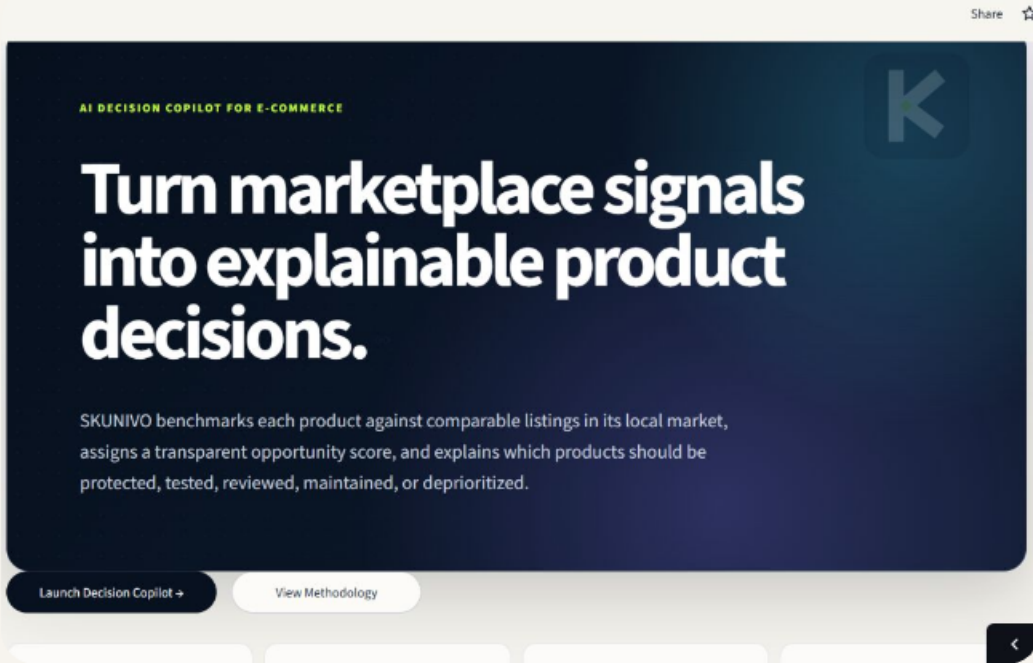
PILOT MEASURES

- Time-to-triage versus week-one baseline
- Recommendation accept / override / defer rate
- Explanation usefulness and user trust
- Number of testable hypotheses created

Financial impact becomes measurable after transaction, cost and experimental data are connected.

The MVP is live today and the architecture can scale in layers

The pilot uses precomputed outputs; future phases can connect APIs and outcome data without redesigning the decision workflow.



LIVE NOW

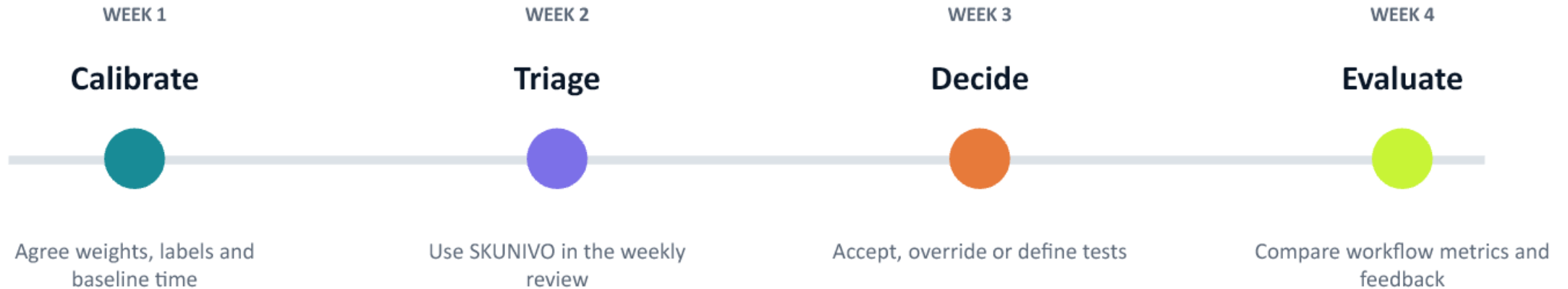
- Public Streamlit application
- English and Vietnamese interface
- Decision Log + Google Sheet feedback

SCALE NEXT

- Scheduled marketplace ingestion
- Warehouse/API integration and access control
- Outcome monitoring and model governance

Four weeks are enough to validate adoption, efficiency and trust

Each stage ends with evidence; automation only expands after the team passes the gate.



PROPOSED SUCCESS GATES

≥70% decisions logged · ≥25% lower median triage time · ≥4/5 explanation usefulness · zero critical data-quality failures

Targets are pilot criteria—not observed results—and should be confirmed with the operating team.



THE DECISION WE ASK FOR

**Approve one market.
One team. Four weeks.**

PILOT

Explainable prioritization

NEXT

**Add transactions + test flags
for lift measurement**

SCALE

**Add costs + inventory
for profit and stock decisions**

SKUNIVO should grow only when the evidence grows.

This prototype prioritizes products using observed marketplace signals. It does not estimate causal promotion lift or forecast transactional demand with the current three-day snapshot dataset.